GEOTECHNICAL INVESTIGATION

Proposed Medina Villas
F.M. Highway 544
Plano, Texas

Project No. 11-DG5327

Prepared for:

MAJESTIC CAST, INC.
Garland, Texas

Prepared by:

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December 22, 2011

Project No. 11-DG5327

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Majestic Cast, Inc.
1625 Ferris Road
Garland, Texas 75044

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Geoscience Engineering & Testing, Inc. is pleased to submit this geotechnical investigation for the above referenced project located in Plano, Texas. This study was authorized by Mr. Sammy Jibrin with Majestic Cast, Inc.

Our Construction Materials Testing Division can provide the materials testing services that will be required during the construction phase of this project. We will be pleased to discuss a scope of work and submit a proposal for these services upon request.

We appreciate the opportunity to be of assistance on this project. Please feel free to contact us if you have any questions or if we can be of further service.

Very Respectfully,

Geoscience Engineering & Testing, Inc.
Firm Reg # F-11285

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INTRODUCTION

Project Description

This report presents the results of the geotechnical investigation performed at the site of the referenced project located in Plano, Texas. Based on the information provided, it is our understanding that a residential subdivision consisting of 45 one-to-two story single-family residences collectively referred to as "Medina Villas" is proposed for construction at this site. Information regarding structural loads was not available at the time of this investigation; however, we anticipate the loads will be from light to moderate. It is expected that the finished floor elevations of the proposed residences will be above surrounding ground surface. A copy of the grading plans were provided to our firm by the client. No other information regarding the referenced project was provided.

Site Description

The site of the referenced project is located along the south side of F.M. Highway 544 and east of Star Court in the City of Plano, Texas. At the time of this study, the site was a vacant tract of land covered with native vegetation. Based on a review of the topography map provided to our Firm by the client, the site slopes gently downwards towards south, however, an abrupt slope in the northern section of the site was also noted. The general location and orientation of the site is shown in the **Illustrations** section of this report.

Purposes and Scope of Work

The principal purposes of this investigation were to evaluate the general soil conditions at the referenced site and develop recommendations for the design and construction of the proposed Apartment Buildings. These purposes were accomplished by:

1). developing subsurface soil and rock stratigraphy at the boring locations;

2). evaluating soil swell potential and alternatives to reduce the soil movement;

3). providing recommendations for foundation design parameters;

4). providing pavement recommendations and

5) providing site preparation recommendations.

Report Format

The first sections of this report describe the field and laboratory phases of the study. The remaining sections present our engineering analyses and geotechnical parameters for the type of foundation system proposed for use at this site. Boring logs and laboratory test results are presented in the **Illustrations** section of this report.

FIELD INVESTIGATION

A total of thirteen (13) test borings were drilled and sampled to a depth of 20 feet from existing ground surface elevation using a truck mounted drill rig. The approximate locations of the borings are shown on the Boring Location Plan Plate A. Logs of the borings with descriptions of the soils sampled are presented on Plates 1 through 13. Soil strata boundaries shown on the boring logs are approximate.

The borings were advanced using continuous flight augers. Undisturbed cohesive soil samples were obtained using a 3-inch diameter thin-walled tube sampler pushed into the soil. The undrained compressive strength of cohesive soils was estimated in the field using a calibrated pocket penetrometer.

To evaluate the relative density and consistency of harder formations, Texas Department of Transportation Cone Penetrometer tests were performed at selected locations. The actual test consists of driving a three-inch diameter cone with a 170-pound hammer freely falling 24 inches. In relatively soft materials, the penetrometer cone is driven one foot and the number of blows required for each six-inch penetration is tabulated at respective test depths, as blows per six inches on the boring logs. In hard materials, the penetrometer cone is driven with the resulting penetrations, in inches, accurately recorded for the first and second 50 blows for a total of 100 blows. The penetration for the total 100 blows is recorded at the respective testing depths on the boring logs.

All soil samples were removed from the samplers in the field, visually classified and wrapped in plastic bags to prevent loss of moisture or disturbance during transfer to the laboratory. The borings were drilled using dry auger procedures to observe water level at the time of the exploration. These water level observations are recorded on the boring logs.

LABORATORY TESTING

Engineering properties of the foundation soils were evaluated in the laboratory by tests performed on representative soil samples. A series of moisture content and dry unit weight determinations were

performed to develop soil moisture profiles and to aid in evaluating the uniformity of soil conditions at this location. Dry unit weight determinations, Liquid and Plastic limit tests (Atterberg limits) and percent passing number 200 sieve were performed on selected soil samples to confirm visual classification and evaluate soil volume change potentials.

Shear strengths of cohesive soils were estimated in the field and in the laboratory by pocket penetrometer tests performed on selected soil samples. All tests were performed using following ASTM procedures.

Laboratory Test	Applicable Test Standard
Liquid Limit, Plastic Limit, and Plasticity Index of Soil	ASTM D 4318
Moisture Content	ASTM D 2216
Unconfined Compressive Strength of Cohesive Soils	ASTM D 2166
Percent Passing No. 200 Sieve	ASTM D 1140

Review

Descriptions of strata made in the field were modified in accordance with laboratory tests results and visual examination in the laboratory. All recovered soil samples were examined, classified and described in accordance with ASTM D 2487, ASTM D 2488 and Unified Soil Classification procedures. Classifications of the soils and finalized descriptions of soil strata are shown on the attached boring logs.

GENERAL SUBSURFACE CONDITIONS

Area Subsurface Stratigraphy

According to the Bureau of Economic Geology, Geologic Atlas of Texas, the property lies within the “Austin Chalk” geological formation.

Typically, "Austin Chalk" (Kau) geological formations consist of; 1) Upper and lower parts, chalk, mostly microgranular calcite, massive, some interbeds and partings of calcareous clay, thin bentonitic beds locally in lower part, forming westward-facing scarp; light gray and 2) Middle part, mostly thin-bedded marl, with interbeds of massive chalk, locally burrowed, marcasite-pyrite nodules common, light gray, weathers white, marine megafossils scarce, thickness 300-500 feet, thins southward.

Local Subsurface Stratigraphy

Based on our interpretation of the borings drilled for this study, the general subsurface stratigraphy encountered at this site consists predominately of clay underlain by calcareous clay deeper underlain by limestone.

The stratigraphy encountered in the test borings drilled for this project consisted of:

Stratum I —— Dark brown CLAY (CH)** from existing ground surface elevation to a depth of 1 to 6 feet

Stratum II —— Tan and brown CLAY (CH)** soils were encountered below stratum I and remained visible to the depths of 3 and 8 feet.

Stratum III —— Tan CALCAREOUS CLAY (CL)** soils were encountered below stratum II and remained visible to the depths of 4 and 9.5 feet.

Stratum IV* —— Tan weathered LIMESTONE was encountered below stratum III and remained visible to the completion depths of the borings drilled.

* Gray limestone was encountered below tan limestone in the test boring B-13 at a depth of 18 feet.
** As per Unified classification System

Detailed descriptions of the subsurface stratigraphy encountered at the locations of the test borings drilled for this study are included on Plates 1-13 in the **Illustrations** section of this report.

Subsurface Water Conditions

All the borings were advanced using dry auger drilling methods in order to observe any groundwater seepage levels. Ground water seepage was NOT encountered in any of the test borings drilled for this site.

The depth to subsurface water will be affected by changes in atmospheric conditions and site drainage. With regard to the aforementioned, we recommend the depth to groundwater be verified during construction. Any noticeable variations in the conditions reported herein should be conveyed to this office immediately so that a thorough review of the current design recommendations can be made. Based upon short-term observations, it is not possible to accurately predict the magnitude of subsurface water fluctuations that might occur. In addition, it is not uncommon to detect seasonal ground water at the interface of limestone and clay particularly after periods of heavy rains.

ANALYSIS AND RECOMMENDATIONS

Potential Vertical Soil Movement

The near surface clay soils encountered within the test borings exhibited plasticity indices of 24 to 54 which are considered as moderately to highly expansive in nature. The clay soils encountered are capable of vertical movement with changes in moisture conditions. The magnitude of the moisture induced vertical movement was calculated using the Texas Department of Transportation (method 124-E). Based on the aforementioned method, in conjunction with the current moisture profile the estimated moisture induced vertical movement or potential vertical rise (PVR) of the soil at existing ground surface of the borings drilled for this study is on the order of 3 to 4.25 inches. Considerably more movement will occur in areas where water ponding is allowed to occur during and/or after construction. Site grading and thickness of the clay soils from the finished grade can also increase/decrease the potential vertical rise.

FOUNDATION DESIGN CRITERIA

A: Pier Type Foundation System

The structural loads can be supported by auger excavated, straight-sided, cast-in-place, reinforced concrete piers. The piers should be founded 6 feet within tan limestone encountered at depths of 4 to 9.5 feet below ground surface at the test boring locations. The net allowable end bearing capacity of 12,000 psf and skin friction of 1,500 psf can be used. The skin friction component should only be applied to the portion of the shaft located in the bearing material below the recommended minimum penetration.

The piers should be provided with enough steel reinforcement to resist the uplift pressures that will be exhibited by the near surface soils. The uplift pressure due to clay soils can be approximated on the order of 1,800 pounds per square foot of shaft area over an average depth of 8 feet or top of limestone whichever is encountered first. Foundation piers designed and constructed in accordance with the information provided in this report will have a factor of safety in excess of 3 against shear type failure and will experience minimal settlement (less than one inch).

Pier Installation

The construction of all piers should be observed by experienced geotechnical personnel during construction to ensure compliance with design assumptions and to verify: (1) the bearing stratum; (2)

the minimum penetration; (3) the removal of all smear zones and cuttings; (4) that groundwater seepage, if encountered, is correctly handled; and (5) that the shafts are vertical and within the acceptable tolerance levels.

Reinforcing steel and concrete should be placed immediately after the excavation has been completed and observed. In no event should a pier excavation be allowed to remain open for more than 8 hours. Concrete should be placed in such a manner as to prevent segregation of the aggregates. Subsurface conditions at the time our borings were advanced indicate that temporary casing will not be required. However, based on our prior experiences with other projects performed within the vicinity of this site, casing of some of the foundation piers may be required after periods of heavy rain.

Grade Beams

Grade beams should be structurally connected into the top of the piers. If a ground supported grade beam design is planned then the estimated soil vertical movement should be reduced to an inch by adopting ay of the method outlined in this report.

Alternatively, a minimum void space of 6 inches should be provided beneath the grade beams. As mentioned earlier that in the event fill soils improvement is not planned then the structure should be designed on void box (suspended). This void space allows movement of the soils below the grade beams without distressing the structural system. Structural cardboard forms are typically used to provide this void beneath grade beams. Cardboard forms must have sufficient strength to support the concrete grade beams during construction.

Floor Systems

Two types of floor systems may be considered for use at this site:

i). **Suspended Floor System** - The floor system of the proposed building should be structurally supported on the foundation piers and a minimum void space of 6 inches should be provided between the bottom of the slab and the underlying soils. In the event this system is used then fill soils improvement is not required.

ii). **Ground Supported Slab** - In conjunction with drilled shafts, a ground-supported slab may be considered for use at this site, providing the risk of some post-construction movement is acceptable.

A ground-supported slab, if used, the slab should be designed to resist the anticipated vertical movement at this site or the soil swell potential should be reduced to one inch by the following methods:

- *By Placement of select fill soils*:

In the event that a reduction in the vertical movement is planned, then the existing highly expansive soils should be replaced with select fill soils.

The reduction in the vertical movements after the placement of select fill soils under controlled conditions is as follows:

Thickness of Select Fill (feet)	PVR Movements (± inches)
0	4.25
2.5	3
4	2
5.5	1

Placement of expansive soils on the building pads will increase the potential for vertical movement of the soils at this site; therefore, we recommend using the specification of the select fill outlined in the **Select Fill Section** of this report.

- *By moisture conditioning of the sub-grade soils:*

Remove the subgrade soils to the top of the limestone from the finished grade elevation and stockpile. The previously removed fill soils should be placed back in the building pad area in 6 to 8 inch loose lifts and each lift should be compacted to between 93 and 98 percent of the maximum standard proctor dry density with a minimum moisture content of 4 points above optimum. The upper one foot of soils in the building pad area should consist of select fill soils placed on the moisture conditioned subgrade.

We recommend the improvements extend an additional five feet beyond the perimeter of the building pad. However, the upper one foot of the select fill soils should not be extended beyond the building line however, should be capped with on site highly plasticity index clay soils in order to resist any seepage of the water in the underlying subgrade soils.

Building Pad Preparation

All existing surface vegetation, loose fill and debris should be removed. The exposed surfaces should then be scarified, watered as required and re-compacted to between 95 and 100 percent of the maximum dry density as defined by ASTM D 698 (Standard Proctor Test). Moisture contents for the existing soils should be within four percentage points of optimum value. If a suspended floor system is planned, the site may then be filled to grade using a suitable fill, free from deleterious matter. If use of a ground supported floor system is intended - or – if a reduction in the vertical soil movement is planned, then the use of select fill soils –or – moisture conditioning of the subgrade soils is recommended. Moisture conditioning procedures are outlined in a previous section of this report. The amount of select fill required to reduce the vertical soil movement to a tolerable level is outlined in the **Floor System** Section of this report. Fill material should be placed in six (6) to eight (8)-inch loose lifts with a moisture content between optimum and four percentage points above optimum (within three percentage points for select fill soils). Each lift should be compacted between 95 and 100 percent of the maximum dry density as defined in ASTM D 698. Field density tests should be taken at the rate of one test per every 2,500 square feet per lift, or a minimum of 3 tests per lift in the area of all compacted fill. For areas where hand tamping is required, the testing frequency should be increased to approximately one test per lift, per 100 linear feet of area.

B. Slab-on-Grade Type of Foundation System

Slab- on- Grade type of foundation system is feasible for use at this site provided:

- The foundation is designed to resist the anticipated soil vertical movement at the site –or- soil potential vertical rise is reduced to more tolerable limit by choosing one of the methods outlined in this report.
- the risk of some post-construction movement is acceptable. Slab on grade type of foundation system require periodic cosmetic repairs to building finishes and similar maintenance. Additionally, there is a risk that the ground movements could be greater than anticipated, which could lead to the need for more extensive repairs. These circumstances are most often associated with poor drainage around the slab perimeter, sub-slab plumbing leaks, or trees or shrubs planted too close to the foundation. The potential for these movements, risks associated with long-term performance of slab-on-grade foundations, and owner maintenance responsibilities should be fully understood.

The proposed residential building(s) can be supported on a structurally designed stiffened slab-on-grade type foundation. The slab may be a grid-type grade beam and slab reinforced with post-tension tendons or conventional rebars. The foundation should be designed with exterior and interior grade beams adequate to provide sufficient rigidity to the foundation system to sustain the vertical soil movements expected at this site. The minimum depth of grade beam should be 2 feet, however, to have an effect of vertical barrier to the edge and center moisture variation, VOLFLO recommends a minimum of 2.5 feet of vertical barrier. Grade beam can act like a vertical barrier. A net allowable soil bearing pressure of 1,500 pounds per square foot may be used for design of all grade beams and slab bearing in natural clay density controlled or moisture conditioned fill.

Grade beams should be designed by the structural engineer to sustain the movements expected at this site. The beams should be founded within natural soils or recompacted and tested fill. The bottom of the beams should be free of any loose or soft material prior to the placement of the concrete. All grade beams and floor slabs should be adequately reinforced to minimize cracking as normal movements occur in the foundation soils. Also, a moisture barrier of polyethylene sheeting or similar material should be placed between the slab and the subgrade soils to retard moisture migration through the slab. It should be understood by all parties that a soil-supported foundation system will experience movement with time, particularly at this site because of the highly expansive clay soils.

PTI PARAMETERS

In addition to the TxDOT method, another method for calculating vertical soil movements uses procedures outlined by the Post-Tensioning Institute (PTI).

To calculate the "PTI" values, the following assumptions were made and the PTI Manual & engineering judgments were used:

- Horizontal Barrier: None
- Constant Suction: 3.6 pf
- Fabric Factor: 1.
- Maximum Depth to Constant Suction: 7 feet
- Effective plasticity index is : 50
- Edge moisture variation distance for center lift (e_m): 6.1 feet
- Edge moisture variation distance for edge lift (e_m): 4.8 feet

- subgrade coefficient of 0.75 (polyethylene sheeting over sand)
- Allowable bearing pressure: 1,500 psf for natural soils and 1,500 psf for compacted, improved and tested fill

Based on the aforementioned assumptions and engineering judgments, the estimated PTI differential movements (y_m) are as follows:

	DIFFERENTIAL SWELL (y_m) (inches)	
CONDITION	**Center Lift**	**Edge Lift**
For existing soils condition	2.76	3.36
With 4 feet of select fill $e_{m\ edge}$: 4.2, $e_{m\ Center}$: 8.3	1.31	1.81

Differential ground movements of greater magnitude could occur if unusually high moisture variations are allowed to occur such as full soil saturation due to water ponding or extreme soil desiccation during summer droughts.

Slab-on-grade foundations should be designed for the most critical criteria of either differential movements estimated by the PTI procedures or total potential movements (PVR) estimated using the Texas Highway Department Test Method Tex-124-E.

Building Pad Preparation

Prior to placing any fill material, all existing surface vegetation and trees should be removed and disposed of off-site. Tree roots should be excavated and removed. The disturbed areas where trees are excavated from should be widened and deepened until hard stratum is encountered. Then, on site soils should be used to back fill the disturbed area in 6 to 8 inches loose lifts and compacted with 95 to 100 percent with moisture content between optimum and 4 points above optimum. The exposed surface from other areas should be scarified to a minimum depth of 6 inches, watered as required and compacted to 95 or greater of maximum dry density with moisture content between optimum and 4 points above optimum. If additional fill if required, then select fill soils should be

used. If a reduction in the vertical soil movement is planned, then moisture conditioning of the subgrade soils –or- water pressure injection of the subgrade soils should be used. Procedures for moisture conditioning of the subgrade soils are outlined in previous sections of this report. Select fill material should be placed in six (6) to eight (8)-inch loose lifts with moisture content between optimum and three percentage points above optimum. Each lift should be compacted between 95 and 100 percent of the maximum dry density as defined in ASTM D 698. Field density tests should be taken at the rate of one test per every 2,500 square feet per lift, or a minimum of 3 tests per lift in the area of all compacted fill. For areas where hand tamping is required, the testing frequency should be increased to approximately one test per lift, per 100 linear feet of area.

Select Fill

"Select fill," as referred to in this report, should consist of clayey sands free of organic materials and have a Plasticity Index between 8 and 18, a Liquid Limit of 35 or less, and between 25 and 45 percent passing a No. 200 Sieve. The Placement and compaction of select fill should be performed in accordance with the "**Building Pad Preparation**" section of this report.

PAVEMENT AND SUBGRADE RECOMMENDATIONS

It is our understanding that the Client prefers to utilize the existing subgrade soils in the paving area "as is" with no modifications to the soils being made. Based on the soil samples obtained from the test borings drilled in the paving area at this site, in addition to a review of the grading plan provided to us by the client, we anticipate clay or calcareous clay will be exposed after final grading is completed. Based on these factors, the pavement can be constructed on the prepared natural subgrade PROVIDED modest drainage is implemented at the time of construction and maintained throughout the usage span of the paved areas.

The following concrete pavement sections are recommended for use at this site:

	Thickness (inches)
Heavy Traffic	
Portland Cement Concrete	8*
Compacted Subgrade	6

*design is based on:

- ESAL's: 100,000
- Reliability level: 85 percent
- Combined standard error : 0.35
- Initial serviceability index of 4.5
- Terminal serviceability index of 3
- Modules of rupture (psi) : 500
- Drainage factor 0.6
- Load transfer Coefficient 3
- Modulus of subgrade reaction(pci): 10

Prior to the placement of any fill in the pavement area, we recommend the entire paving area be prepared as per the procedures outlined in the **Building Pad Preparation** Section of this report.

If additional fill is required, then on-site soils should be used. The fill should be placed as per the procedures outlined in the **Building Pad Section** of this report.

Proper drainage away from the pavement is the utmost importance to reduce water infiltration into the subgrade. Water ponding should not be allowed to occur on or close to the pavement surface. The street should be raised to avoid any flooding to occur and street embankment should be protected by riprap. Some differential movement in the pavement is anticipated over time due to the swelling of the subgrade clays at this site. Design of the concrete pavement should specify a minimum 28-day concrete compressive strength of 3500 psi with 4 percent to 6 percent entrained air. The concrete should be placed within one and one-half hours of batching. During hot weather, the concrete placement should follow ACI 311 Hot Weather concreting and in no case should the concrete temperature be allowed to exceed 95ºF. To avoid excessive heat periods, consideration should be given to limiting concrete placement to a time of day that will minimize large differences in the ambient and concrete temperature.

Past experience indicates that pavements with sealed joints on 15 foot spacings, cut to a depth of at least one-quarter of the pavement thickness, generally exhibit less uncontrolled post-construction cracking than pavements with wider spacings. As a minimum, expansion joints should be used wherever the pavement abut a structural element subject to a different magnitude of movement, e.g., light poles, retaining walls, existing pavement, building walls, or manholes. After construction, the construction and expansion joints should be inspected periodically and resealed, if necessary. The pavement should be reinforced using at least No. 3 bars, 18 inches on center, each way.

ADDITIONAL DESIGN AND CONSTRUCTION CONSIDERATIONS

1. Positive drainage of the surface water around the foundations is of utmost importance at this site because of the highly expansive nature of the subsurface soils. Downspots should discharge water away from the foundation in order to prevent accumulation of excessive water next to the foundation. The ground surface around the foundations should be sloped downward and away from the residences on all sides.

2. Considerable changes in moisture content of the subsurface soils below the foundations will result in excessive differential foundation movements. Therefore, the homeowner must ensure that the moisture content of soils around the foundation is maintained at a relatively constant level. One method of ensuring that adequate watering is accomplished is by observing the soils next to the foundation periodically (during hot and dry periods in particular) to ensure that soil is not separating or pulling back from the foundation.

3. We recommend that any sprinkler system constructed for the Apartments at this site be placed at least 3 to 5 feet away from the foundation to limit accumulation of excessive water next to the foundation. The sprinkler system, if constructed, should cover the entire perimeter of the residences (grass covered areas) to provide uniform moisture around the foundation in order to limit differential movements.

4. We recommend that during the landscaping of the lots, trees be planted as far from the foundations as possible. This is due to the drastic moisture fluctuations that roots can cause under the foundations. Also, we recommend that selected plants and trees be used during the landscaping. The selected plants and trees should be of the kind that will have a lesser demand for water and will not have vigorous root growing characteristics. Additionally, we recommend that the trees not be planted closer than their mature height from the residences.

5. Although any slab constructed on expansive soils will some experience vertical movement due to seasonal moisture variations, we recommend that the builders and homeowners be advised of the importance of the above conditions and recommendations in order to limit foundation movements.

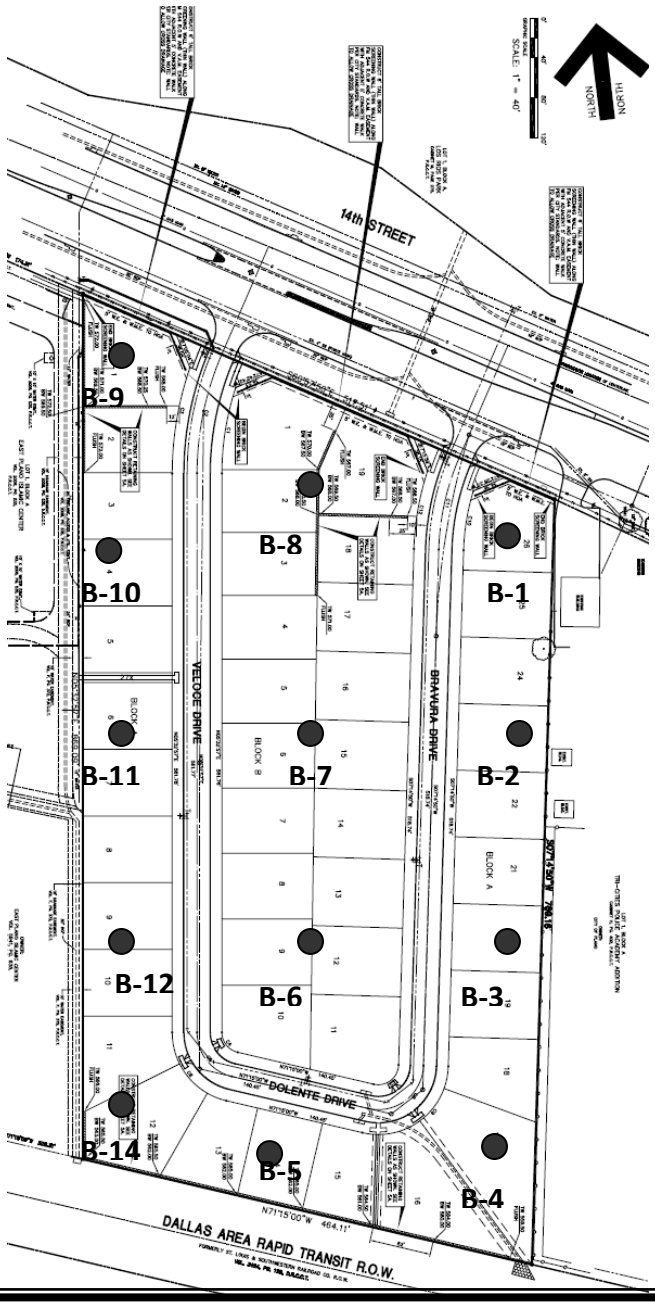
CLOSURE

Soil and moisture conditions may vary between boring locations. Statements in this report as to subsurface variation over given areas are intended only as estimations from the data obtained at specific boring locations.

The professional services that form the basis for this report were conducted using the degree of care and skill ordinarily exercised, under similar circumstances, by reputable geotechnical engineers practicing in the same locality. No other warranty, expressed or implied, is made as to the professional advice set forth.

The results, conclusions, and recommendations contained in this report are directed at, and intended to be utilized within the scope of work contained in this report. This report is not intended to be used for any other purposes. *Geoscience Engineering and Testing, Inc.*, makes no claim or representation concerning any activity or condition falling outside the specified purposes to which this report is directed, said purposes being specifically limited to the scope of work as defined in said agreement. Inquiries as to said scope of work or concerning any activity or condition not specifically contained therein should be directed to *Geoscience Engineering & Testing, Inc.* for a determination and, if necessary, further investigation.

ILLUSTRATIONS



● Approximate Boring Location

BORING LOCATION PLAN
Proposed "Medina Villas"
F.M. Highway 544
Plano, Texas

GETI Project No. 11-DG5207

Plate A